

Mohammed Abdullah

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SUMMARY

AI Engineer with experience in building both production and proof of concept AI systems across distributed streaming analytics, computer vision, vision-language models, multimodal AI, and machine learning infrastructure. Experienced in FastAPI, Kafka, PyTorch, React, and modern LLM/VLM applications, with hands-on experience processing hundreds of millions of records and deploying real-time AI systems.

EDUCATION

M.Tech IT (AI & DS) — First Rank Holder

Aug 2023 – May 2025

Anna University, CEG Campus, Guindy | CGPA: 9.0

B.E Computer Science and Engineering — Merit Holder

May 2019 – Apr 2023

Panimalar Engineering College, Affiliated to Anna University | CGPA: 9.51

TECHNICAL SKILLS

Languages: Python, C++, TypeScript, JavaScript, SQL

AI / ML: PyTorch, LangChain, LangGraph, LangSmith, Milvus, FAISS Vector Search, Transformers, Large Language Models (LLMs), Vision-Language Models (VLMs), Huggingface TRL, Scikit-Learn, MLflow, AMD Vitis HLS

Computer Vision: OpenCV, MediaPipe, YOLO, CLIP Embeddings from Hugging Face Transformer, Object Detection, Pose Estimation, Gaussian Splatting, Photogrammetry

Systems / Data: Apache Kafka, Redis, Celery, Parquet, PyArrow, RAPIDS cuDF, SQLite

Backend: FastAPI, Flask, Node.js, REST APIs

Frontend: React vite, TypeScript, HTML, CSS

Tools / Platforms: Git, Linux, Docker, Ollama, TensorBoard, Unity ML-Agents, Matplotlib, Seaborn

EXPERIENCE

AI Engineer 1

Sep 2025 – Present

BBBS Pvt. Ltd. – Defence Technology Solutions, Chennai

Agentic AI + Deep Learning [Anomaly Detection, Time Series Model], Synthetic Data Generation + Scalable Distributed System

Maritime Vessel Analytics and Anomaly Detection Platform | *Python, PyTorch, FastAPI, React, langchain, langgraph, Docker*

- Built a distributed maritime analytics platform processing **244M+ AIS vessel records** for vessel tracking, route analysis, and anomaly detection across global shipping traffic.
- Developed FastAPI services and React-based dashboards enabling real-time vessel tracking (via APIs), trajectory visualization (current stored data and real-time fetched data), proximity analysis (region based for multiple vessels), operational investigations (anomaly detection based on vessel movement), and event-driven anomaly alerts (Using Navtex and Navarea warnings from Hydrobharat feeds).
- Designed and implemented a streaming AIS analytics architecture using Apache Kafka, Redis, Celery, and Docker, supporting scalable real-time ingestion, distributed processing, task orchestration, caching, and anomaly-detection workflows.
- Designed resilient event-driven data pipelines with Kafka, including **Dead Letter Queue (DLQ)** mechanisms for fault isolation, message replay, and reliable processing during schema evolution and pipeline failures. Validated pipeline resilience using poison-pill message testing and DLQ-based recovery.

- Developed multi-step vessel trajectory forecasting models to predict up to **10 future AIS positions** by encoding vessel locations as Geohash representations and decoding model outputs back to geographic coordinates. Fine-tuned AutoGluon-based models that achieved an **11% reduction in geospatial position Haversine distance error** compared to ARIMA and Prophet baselines on the Piraeus AIS dataset. Leveraged trajectory deviations, AIS transmission gaps, and movement inconsistencies as signals for abnormal vessel behavior detection.
- Generated 6.8M+ synthetic AIS spoofing trajectories (2.8% of a 244M-record dataset) using mathematical trajectory models and LLM-guided augmentation to simulate anomalous vessel behavior for anomaly detection.
- Designed a GPU-accelerated maritime analytics pipeline leveraging **Parquet** partitioned datasets for columnar storage, **PyArrow Dataset API** for predicate pushdown and selective column loading, and **RAPIDS cuDF** for GPU-based joins, aggregations, trajectory feature extraction, and geospatial filtering across **244M+ AIS records**. Achieved sub-**3 second** retrieval of **20K vessel records** within user-selected map regions and time ranges, supporting real-time investigation workflows through Celery-managed tasks and Leaflet-based visual analytics.
- Improved anomaly assessment by correlating vessel telemetry with environmental conditions (weather datasets, maritime regulations (land domains from NOAA Maritime Zones and Boundaries dataset), and historical movement behavior (AIS data mentioned above) to reduce false-positive investigations.

Edge AI + FPGA Accelerator Architecture + RTL Hardware/Software Co-Design

Real-Time YOLOv8-Nano FPGA Inference Accelerator for UAV Object Detection | *Python, C++, Vitis HLS, Verilog, Vivado, PyTorch, Ultralytics YOLOv8*

- Designed a reusable weight-stationary FPGA accelerator for the complete **23-layer YOLOv8-Nano** network targeting the Xilinx Kria KV260, replacing a monolithic implementation with serialized DDR-backed execution using reusable compute engines.
- Implemented an end-to-end hardware inference pipeline converting real UAV video into FPGA-ready tensors, executing RTL simulation, decoding detection logits, performing confidence filtering and NMS, and validating detections against the PyTorch YOLOv8 reference pipeline.
- Built an automated PyTorch-to-RTL deployment flow by extracting and quantizing **1.88M detection-head parameters** to INT8, reducing weight storage from **6.5 MB to 1.7 MB (74% reduction)**, and generating synthesizable Verilog with embedded weights.
- Implemented streaming HLS kernels for Convolution, C2f, Bottleneck, SPPF, Upsample, Concat, and Detect modules with reusable processing elements, shared requantization pipelines, and serialized layer scheduling to minimize FPGA resource utilization.
- Performed **11 FPGA architecture experiments** exploring streaming execution, DDR tiling, compute parallelism, memory hierarchy, RTL timing, and buffer reuse to derive the final accelerator architecture.
- Reduced FPGA resource utilization through architectural redesign, lowering **BRAM usage by 99.9% (15,296→11)**, **DSP utilization by 95.4% (1348→62)**, **LUT usage by 60.1% (231K→92K)**, and **flip-flops by 53.3% (115K→54K)**, enabling complete deployment on the KV260.
- Validated RTL functionality using automated regression testing, HLS C simulation, RTL co-simulation, and Python reference comparisons, achieving bit-exact layer verification and less than **0.2% latency deviation** between synthesis estimates and RTL simulation.
- Generated synthesizable Vivado IP and integrated the accelerator into a Zynq UltraScale+ KV260 block design, successfully completing synthesis, place-and-route, and bitstream generation while achieving **timing closure (Worst Negative Slack = 0.00 ns)** for deployment.

Vision-Language + Computer Vision + Activity/Behavior Recognition + Multimodal AI + Retrieval (FAISS)

POC: Behavior Inference System for Industrial Assembly | *Python, VLMs, MediaPipe, YOLO, FAISS*

- Developed a real-time worker activity recognition system for industrial assembly environments using computer vision and multimodal AI pipelines, for Proof of concept for internal application.
- Benchmarked Vision-Language Models including Qwen2-VL, Florence-2, InternVL, OpenFlamingo, and BLIP, in internal dataset for inference speed, contextual understanding, and deployment constraints.
- Designed MediaPipe and YOLO-based spatiotemporal feature extraction pipelines to model worker posture, object interactions, motion patterns, and task progression.
- Implemented temporal evidence aggregation, task-subtask retrieval, and FAISS-based context retrieval workflows, improving behavioral classification performance from **78.4% to 86.1% Macro-F1** while reducing spurious frame-level label transitions by **62%**.

- Built FastAPI backend services, Server-Sent Events (SSE) streaming infrastructure, and analytics dashboards for real-time monitoring and behavioral statistics reporting.

Computer Vision + GPU Optimization + Profiling

Photogrammetry and Gaussian Splatting Research | *Gaussian Splatting, CuBLAS + CuDNN, Nsight Compute*

- Built and maintained a multi-pipeline 3D reconstruction platform integrating COLMAP (which is SfM) from Nerfstudio for sparse reconstruction of 3D point cloud from 2D images using Optimization of Cholesky function with covariance matrices of feature mapped points between images, and dense reconstruction (which is MvS) for generation of 3D scene with Deep learning.
- Benchmarked Nerf models, ReSplat, DepthSplat, Instant-NGP, for Gaussian Splatting output obtaining 25 peak PSNR in internal dataset for splatfacto model of Nerfstudio; which are the dense reconstruction phase of MVS phase. Also, looked at alternates for Meshroom -AliceVision, and researched on their Feature Matching using AVX and SSE optimizations and why CUDA in this case would be a bad choice and also had benchmarked the same internally created datasets in RealityCapture for reference and studied it's process in a black box manner.
- Used Nsight Compute to identify covariance-computation CUDA kernels as a dominant bottleneck in the feature-matching pipeline; optimized covariance estimation for Cholesky-based optimization for improved GPU parallelism, reducing reconstruction time by **3–5 minutes** on multiple 10-second 1080×1920 indoor video sequences.
- Implemented adaptive GPU memory management with automatic VRAM detection and parameter scaling, enabling reconstruction pipelines to run reliably across RTX 4060 (8GB) through RTX 4090 (24GB) GPUs.
- Resolved inference OOM failures by optimizing memory-intensive reconstruction stages, reducing peak GPU memory consumption from **9.6GB to 6.3GB** through BF16 aggregation, cache-window tuning, and pipeline-level memory management.

Reinforcement Learning + Visualizations

Multi-Agent Pursuit–Evasion Simulation | *Unity ML-Agents, PPO, Self-Play, Plotly, MLflow*

- Contributed to the development of a multi-agent reinforcement learning environment in Unity ML-Agents using PPO, self-play, and curriculum-learning strategies; designed reward functions governing pursuit efficiency, evasion behavior, trajectory planning, agent coordination, and competitive interactions.
- Conducted large-scale experimentation involving curiosity rewards, replay-buffer configurations, learning-rate schedules, self-play settings, and PPO hyperparameters; managed experiments using MLflow for metric tracking, model versioning, hyperparameter analysis, and reproducible training workflows.
- Developed Plotly-based analytics dashboards with animated 3D trajectory visualization to analyze policy evolution and agent behavior, while building evaluation pipelines to detect reward hacking, policy regressions, unstable value-function learning, convergence failures, and performance degradation across training runs.

PROJECTS

Generative AI + Multi-Agent RAG Systems + Educational AI

Curriculum-Aware Multi-Agent RAG Platform for Personalized Educational Content Generation. | *Python, LangGraph, Ollama, Milvus, SentenceTransformers, FastAPI, DOCX Generation*

- GitHub Repository: <https://github.com/mohammed1916/RAGmultiAgentRevisableDocGenerator>
- Built a curriculum-aware multi-agent RAG platform that generates personalized learning plans and structured educational documents for CBSE Class 10/12 and JEE curricula using curriculum-specific vector databases.
- Designed a LangGraph-based multi-agent workflow for planning, content generation, review, and document generation using Router, Planner, Writer, and Reviewer agents.
- Implemented curriculum-aware retrieval using SentenceTransformer embeddings and Milvus vector search to provide context-specific recommendations from educational knowledge bases.
- Developed a state-aware planning framework that extracts and validates student progress, tracks learned topics and timelines, and dynamically adapts learning plans based on individual progress.
- Built an end-to-end document generation pipeline that converts AI-generated content into structured Markdown and Microsoft Word (.docx) files through automated formatting and review workflows.

- Built an automated RAG evaluation pipeline measuring retrieval and generation quality: **ROUGE-L: 0.68, BLEU: 0.52, semantic similarity: 0.82, groundedness: 0.91, context utilization: 0.78** across curriculum-specific validation sets.

Computer Vision + ML in Kaggle Competition

Multi-Agent Mouse Behavior Recognition | *Python, LightGBM, XGBoost, CatBoost, Scikit-Learn, Polars*

- Engineered **280+ trajectory, kinematic, interaction, curvature, and temporal features** from multi-body-part pose-estimation sequences containing over **12M+ behavioral observations**, supporting both single-agent and pairwise social-interaction behaviors across datasets with **4–18 tracked body parts**.
- Trained action-specific ensemble models using **LightGBM, XGBoost, and CatBoost**, incorporating adaptive thresholding, temporal smoothing, and event-segmentation techniques to convert frame-level predictions into behavior intervals optimized for the competition's **event-level F-score** evaluation metric.
- Used **Pandas** and **Polars** for large-scale feature generation and experimentation, achieving a **0.078 private leaderboard score** and **1190th rank** in the Kaggle Multi-Agent Mouse Behavior Recognition competition.

Computer Vision + Quantization + Multi model Training

Vision Transformer Research and GPU Optimization | *PyTorch, CUDA, Nsight Compute, Tensor Cores, cuDNN*

- Profiled SegFormer-B0 inference on an RTX 4060 using Nsight Compute, identifying memory-bound execution through cache utilization, bandwidth saturation, occupancy analysis, and kernel stall investigation.
- Implemented BF16 mixed-precision inference with cuDNN and Tensor Core acceleration, reducing activation memory footprint and improving arithmetic throughput. Reduced inference latency from **32.01 ms to 22.04 ms (1.45× speedup)** while maintaining segmentation quality with less than **2% mIoU degradation** on the Cityscapes validation set compared to FP32 inference.
- Inspired by SegFormer's Vision Transformer architecture, designed a multi-task model with a shared transformer encoder and dual decoders for semantic segmentation and monocular depth estimation on the Cityscapes dataset, achieving only **8% mIoU degradation** in segmentation and **11% degradation** in depth estimation when training at **256×256** resolution instead of computationally intensive **1024×1024** inputs.

LLM Finetuning in Kaggle Competition

NVIDIA Nemotron Model Reasoning Challenge | *LoRA, Synthetic Data Generation, vLLM, LLM Fine-Tuning*

- Constructed a synthetic reasoning dataset by extracting structured mathematical and logical problem patterns from **AIMO** and **NVIDIA Wonderland** datasets, converting implicit reasoning structures into executable problem templates using GPT-based models for programmatic transformation and answer derivation. Using those templates, generated augmented training samples by creating semantically equivalent reasoning problems with step-consistent solutions in batches of 1000, and validated outputs using **Claude-based verification**, filtering inconsistent samples before storing the final dataset in structured CSV format for fine-tuning. Maintained class balance across arithmetic, algebraic, combinatorics, physics-inspired reasoning, and logic-based problem types to prevent distribution skew during LoRA training.
- Fine-tuned **LoRA adapters (tried different ranks from 32 to 16)** on the **Nemotron-3-Nano-30B** model using vLLM-compatible inference pipelines, improving structured reasoning performance under constrained parameter-efficient adaptation and achieving a **0.840 Kaggle leaderboard score** with **1916th rank**.

CERTIFICATIONS

- **NVIDIA** — **Accelerated Computing** with CUDA Python
- **NVIDIA** — Transformer-Based NLP Applications
- **Microsoft Applied Skills**: Build a natural language processing solution with Azure AI

ACHIEVEMENTS & ACTIVITIES

- **Merit Scholarship** — Awarded by the Government of Tamil Nadu for securing **91.793 percentile** in TANCET.
- **Kaggle Competitor — Top 7% (14/217 teams)** in ESA “Trojan Horse Hunt in Time Series Forecasting”; participated in **10+ Machine Learning and Deep Learning competitions** including NeurIPS Code Golf and ARC Prize.
- Participant in **NVIDIA GPU Accelerated Computing Codeathon** organized with IEEE Computational Intelligence Society and ACM (Oct 2024).
- Participated in multiple hackathons including **Google Cloud, AWS, and MongoDB developer competitions** (2024–2026).